MILITARY SPECIFICATION

RECORDER-REPRODUCER, SOUND (MAGNETIC TAPE),

TYPE RD-219( )/U

1. SCOPE

1.1 This specification covers a multi-speed 4 channel magnetic tape recorder-reproducer designed for recording high speed code. This recorder-reproducer shall be designed for permanent installation and continuous use. The code shall be played back at a reduced speed and shall be reproduced at an approximate frequency of 1000 cycles per second (c.p.s.) independent of the playback speed. The recorder-reproducer shall also be adaptable for recording and replaying voice at any of its available speeds. The equipment shall contain all the electronics necessary for the recording, equalization and playback of 4 simultaneous input signals. It shall have a loudspeaker for playback of anyone of the 4 recorded signals. A microphone shall be supplied with the recorder-reproducer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

W-T-61 - Tapes, Recording, Magnetic Oxide Coated; and Reels, Recording Tape.

MILITARY

MIL-T-27 - Transformers and Inductors (Audio, Power, and Pulse).

MIL-C-62 - Capacitors, Fixed, Electrolytic (DC, Aluminum Dry Electrolytic, Polarized).

MIL-J-641 - Jacks, Telephone, General Specification for.

FSC 5835

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MILITARY (cont'd.)

MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.

MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).

MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

MIL-I-17623 - Interference, Radio, Requirements, Methods and Limits (14 KC to 1000 MC) for Electric Office Machines, Printing and Lithographic Equipment.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-200 - Electron Tubes, and Semiconductor Devices, Diode.

MIL-STD-701 - Preferred and Guidance List of Transistors.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Ave., at 33rd. St., New York 16, N.Y.)

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3. REQUIREMENTS

3.1 Preproduction sample.- A preproduction sample of the magnetic tape recorder-reproducer shall be required for the tests specified in 4.2.1.

3.2 Description.- The recorder-reproducer shall consist of a single unit made up of the following assemblies:

3.2.1 Tape transport and electronic assembly.- The tape transport and electronic assembly shall contain all the mechanical components necessary for transporting tape and all electronic components necessary for amplification and equalization during the record and playback modes. A loudspeaker shall be mounted in this assembly which will reproduce the selected recorded signal.

3.2.2 Enclosure assembly.- The enclosure assembly shall contain terminal boards for connection of ships wiring and contain all the necessary plugs or jacks or both for connection of the tape transport and electronic assembly to the appropriate terminal boards.

3.3 General features.- The equipment shall be in accordance with the following paragraphs of Specification MIL-I-983 in addition to the requirements specified herein: (Whenever a requirement of Specification MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

General information
Definitions
Materials, general
Stressed or vital parts
Substitution of (equal or superior) materials
Fungus-inert materials
Unacceptable materials
Acceptable materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Aluminum
Magnesium
Iron or steel
Brass
Bronze
Copper
Coper-beryllium alloy
Copper-nickel alloy
Nickel-copper alloy (Monel)
Copper-nickel zinc alloy (nickel silver)
Zinc
Other metals
Springs
Plastics
Impregnating, filling, potting, embedding and encapsulating compounds
Lubricants and lubrication

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Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Gaskets
Locking devices
Washers
Ball bearings
Parts - electrical - general
Capacitor sockets
Capacitors
Variable resistors
Transformers
Relays
Electrical tapes
Switches
Fuses
Enclosures - general
Enclosure - accessibility
Enclosure - degree of (dripproof on top)
Stiffening grooves
Minimum sheet metal thicknesses
Through bolting
Ventilation
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity (extreme temperature for exposed location apply)) (see 3.7.1)
Shock, vibration and inclination (see 3.5.3.1)
Primary power supply circuits
Power supply tolerances (see 3.6.9)
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection (test cable required)
Terminal boards and connectors and terminals
Wiring
Color coding
Electrical insulation
Airborne, audible noise
Structureborne noise (vibration) for submarine equipment
Drawings - general
Drawing - preliminary
Drawing - working
Schematic diagrams
Wiring diagram
Drawing list
Assembly drawings
Bill of materials
Interchangeability and standardization
Technical manuals (see 3.10)
Repair parts (electronic)
Designation and marking (see 3.9)
Design for production
Reports
Nomenclature
Workmanship and general examination

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3.4 Functional requirements.-

3.4.1 Time of operation.- This equipment shall be designed for operation over extended periods of time with a minimum of maintenance.

3.4.2 Recording and reproducing medium.- The recording and reproducing medium shall be 1/4 inch magnetic tape wound on a 7-inch diameter plastic reel. The tape and reels shall be in accordance with Specification W-T-61.

3.4.3 Tape speeds.-

3.4.3.1 Fixed speeds.- The equipment shall have two fixed linear tape speeds. These speeds shall be 7-1/2 inches per second (i.p.s.) and 3-3/4 i.p.s. Tape speeds shall not vary more than plus or minus 2 percent over the entire length of a reel of tape.

3.4.3.2 Variable speed.- The equipment shall have the capability of varying tape speed over a range of 3-3/4 i.p.s. to approximately 1 i.p.s.

3.4.3.3 Speed selection.- A two position switch shall be provided on the front panel of the equipment for the selection of the two fixed speeds (see 3.4.3.1) and a single rotary switch shall be provided on the front panel of the equipment to continuously vary the tape speed (see 3.4.3.2). These switches may be combined into a single speed selector control switch.

3.4.4 Simultaneous record and reproduce.- Facilities shall be provided to record and reproduce simultaneously four channels of information.

3.4.4.1 Interchangeability of recordings.- Recordings shall be interchangeable between any and all equipments manufactured in accordance with this specification. Track dimensions, clearances and relative directions of motion and recording shall apply and shall be in accordance with figure 1. Dimensions shall be considered nominal.

3.4.5 Erasure.- The equipment shall provide a means of erasing the recording by a high frequency erase method. Erasing shall automatically occur during the recording process such that the tape is erased prior to reaching the recording head. The erase frequency shall be a minimum of 50 kc. All four tracks shall be erased simultaneously.

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3.4.6 Operational modes.- The following modes of operation shall be available for selection utilizing either pushbuttons or a single rotary switch:

(a) Record
(b) Playback - audio recording
(c) Playback - code recordings
(d) Stop
(e) Rewind
(f) Fast forward

Changing from operational mode to operational mode shall be accomplished electrically.

3.4.6.1 Record mode.- The equipment shall be capable of recording four channels of information. Each channel shall have a separate record amplifier.

3.4.6.1.1 Record lockout.- An interlock shall be provided to prevent inadvertant switching of the equipment into the record mode.

3.4.6.1.2 Record level control.- A single rotary control shall be included on the front panel to provide level control in record mode for each channel.

3.4.6.1.2.1 Record level indicator.- A decibel (db) meter shall be provided on the front panel of the equipment as a recording level indication. Zero db on the meter shall be optimum record level which is 10 db below test record level.

3.4.6.1.3 Record light.- When the recorder is in the record mode, a pilot light shall remain lighted.

3.4.6.2 Audio playback mode.- The equipment shall be capable of reproducing four channels of audio signal information.

3.4.6.2.1 Playback line amplifiers.- Each channel shall have a separate playback line amplifier.

3.4.6.2.2 Playback power amplifier.- A single power amplifier shall be provided. The input to this amplifier shall be selectable between any one of the four channels, and the output of the amplifier shall drive the internal loudspeaker.

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3.4.6.2.2.1 Monitor channel selection.- A five position rotary switch shall be provided on the front panel of the equipment to allow the selection of any one of the four channels of information for monitoring. The fifth position shall be an "off" position. The output from this switch shall be fed to the monitor selector switch (see 3.4.6.2.2.2).

3.4.6.2.2.2 Monitor selection.- A two position switch shall be provided on the front panel of the equipment to allow the selection of either record input or playback output for monitoring. The output of this switch shall be fed to the record level indicator (see 3.4.6.1.2.1) and the playback power amplifier (see 3.4.6.2.2).

3.4.6.2.2.3 Playback level control.- A level control shall be provided to control the output level of the power amplifier. This control shall only affect the level of the internal loudspeaker.

3.4.6.3 Code playback mode.- The capability of reproducing code signals at a pitch of 1000 c.p.s. when high speed code recordings are played back at a maximum speed reduction of approximately 1/7 the recording tape speed shall be provided for channel 1 (see figure 1). High speed code is received at a rate of 100 words per minute and a frequency of 1000 c.p.s. The code tone generating circuit shall "trigger on" by the peak of the first cycle of the played back code output and shall remain "on" no longer than 5 milliseconds after the peak of the final cycle of a reproduced dot or dash. The output of this circuit shall be fed to the power amplifier input (see 3.4.6.2.2).

3.4.6.3.1 Code defeat switch.- A switch shall be provided on the front panel of the equipment to disconnect the code generating circuit from the power amplifier and allow channel 1 to be capable of reproducing audio information.

3.4.6.3.2 Code sensitivity control.- A level control shall be provided on the front panel of the equipment for the amplitude adjustment of the "trigger on" level of the code tone generating circuit.

3.4.6.3.3 Code pitch control.- A rotary switch shall be provided on the front panel of the equipment to vary output pitch of the reproduced code.

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3.4.6.4 Stop mode.- Facilities shall be provided to stop the movement of tape prior to the selection of another operational mode. The switching arrangement shall be so designed that it is impossible to change operational modes without going through the stop mode. When switching modes, the tape shall not break.

3.4.6.5 Fast forward mode.- The equipment shall be capable of completely winding (fast forward) a complete reel of tape at an average speed of at least 300 feet per minute.

3.4.6.6 Rewind mode.- The equipment shall be capable of rewinding a complete reel of tape at an average speed of at least 300 feet per minute.

3.4.7 Power switch.- A power switch shall be provided which will disconnect both sides of the power supply lines. This switch may be combined with the speed control (see 3.4.3.3). Prior approval is required from the bureau or agency concerned.

3.4.7.1 Pilot light.- A pilot light shall be provided and it shall remain when the power switch is in the "on" position.

3.4.8 Flutter and wow.- Flutter and wow shall not exceed 0.3 percent r.m.s. at a tape speed of 3-3/4 i.p.s.

3.4.9 Reel lock.- A locking device shall be provided on each reel mounting shaft to hold the tape reel on regardless of mode of operation. This device shall be permanently attached to the reel shaft.

3.5 Mechanical requirements.-

3.5.1 Size.- The equipment shall not exceed the following dimensions:

Height - 11-1/2 inches
Width - 16 inches
Depth - 12-1/2 inches

All dimensions shall include shock and vibration mounts (if used) and all projections and controls on the body of the equipment.

3.5.2 Weight.- The weight of the equipment shall not exceed 55 pounds.

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3.5.3 Mounting.- The equipment shall be designed for shelf mounting. By using an adapter, the unit shall be capable of rack mounting when specified (see 6.1).

3.5.3.1 Shock, vibration and inclination.- The equipment shall meet the shock, vibration and inclination requirements for non vital equipment (see 3.3). Shock and vibration isolation mounts may be used, if required, between the equipment and the deck. If shock and vibration mounts are used, a grounding strap shall be provided to insure a good conducting path between the enclosure and the mounting base.

3.5.4 Degree of enclosure.- The equipment shall be dripproof on top when mounted on the shelf mounting.

3.5.5 Equipment assembly.- The equipment shall consist of two main assemblies as follows:

(a) A tape transport and electronics assembly
(b) An enclosure assembly

The tape transport and electronics assembly shall be easily removable from the enclosure.

3.5.5.1 Design for servicing.- The recorder-reproducer shall be designed so that servicing of all parts can be easily and quickly performed when the tape transport and electronics assembly is removed from the enclosure. When these assemblies are removed from the enclosure, it shall be capable of satisfactory operation.

3.5.6 Tape threading.- Tape threading shall be as simple as possible. A diagram of the tape threading path shall be shown on the front panel of the recorder.

3.5.7 Record head and playback head mounting.- The record and playback heads shall be individually mounted to the front panel of the equipment. The head mounting shall be designed to provide azimuth and elevation adjustments of the record and playback heads. These adjustments shall be a locking type adjustment.

3.5.8 Tape footage indicator.- A three-digit Veeder-Root or similar type counter shall be connected to the take-up reel to provide an indication of the amount of tape footage used.

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3.5.9 Take-up and supply reel turntables.- Set screws or taper pins shall not be used to secure reel turntables to the take-up and supply motor shafts.

3.6 Electrical requirements.-

3.6.1 Motors.- This equipment shall have 3 motors. One shall drive the capstan, the other two shall be used to drive the tape reels.

3.6.1.1 Drive motor.- The capstan drive motor may be a d.c. motor.

3.6.1.1.1 Speed changes.- If a d.c. drive motor is used, speed changes shall be made by applying different values of d.c. to the motor. The voltage to be rectified shall be obtained from transformer taps for the 3-3/4 i.p.s. and 7-1/2 i.p.s. speeds. The low speeds may be obtained using a series potentiometer.

3.6.1.2 Take-up and supply motors.- Take-up and supply motors may be either a.c. or d.c. motors. If d.c. motors are used, they shall be identical to the capstan drive motor (see 3.6.1.1).

3.6.1.3 Motor brushes.- If a d.c. motor is used, the motor brushes shall have a minimum continuous operating life of 500 hours. The brush replacement shall be convenient and simple to accomplish.

3.6.1.4 Motor materials.- Requirements for materials included in this specification do not apply to electric motors, but the motors are required to meet the temperature, humidity and all other requirements of this specification.

3.6.2 Electrical components.-

3.6.2.1 Capacitors.- The following types of capacitors may be used:

3.6.2.1.1 Electrolytic capacitors.- Electrolytic capacitors in accordance with Specification MIL-C-62 may be used in power supply filter sections and power isolation filters.

3.6.2.1.2 Tantalum capacitors.- Tantalum capacitors shall be used for interstage coupling and signal bypass applications.

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3.6.2.2 Transformers.- Transformers shall be grade 5, class S, life expectancy X in accordance with Specification MIL-T-27. In low level stages, grade 4 transformers may be used. If grade 4 transformers are used, prior approval shall be obtained from the Bureau of Ships.

3.6.2.3 Metallic rectifiers.- Rectifiers shall be of the silicon diode type, and shall be chosen from approved types as shown in Standard MIL-STD-200.

3.6.2.4 Transistors.- Transistors shall be used for all amplifiers and oscillators. Transistors shall be approved types chosen from Standard MIL-STD-701.

3.6.2.5 Relays.- All relays used shall have bifurcated contacts.

3.6.2.6 Potentiometers.- Subminiature potentiometers may be used for internal adjustment controls.

3.6.2.7 Resistors.- Deposited carbon resistors may be used where low noise characteristics are required. Resistors used in power applications may be of the miniature type.

3.6.3 Amplifiers.- Four record amplifiers, four playback line amplifiers and one power amplifier shall be provided.

3.6.3.1 Inputs.- The record amplifiers shall have the following five inputs:

3.6.3.1.1 Microphone input.- A front panel jack, type JJ-034 in accordance with Specification MIL-J-641, shall be provided for a low impedance dynamic microphone to be supplied as part of the equipment. This jack shall be permanently wired to channel 4 record amplifier input.

3.6.3.1.2 Line input.- The input impedance shall be a minimum of 5000 ohms, unbalanced. There shall be one line input for each of 4 channels on the equipment terminal board.

3.6.3.1.2.1 Input signal range.- The equipment shall be designed to record at the test record level with an input signal range of from 0.25 to 15 volts on the line input of each channel.

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3.6.3.2 Outputs.- The equipment shall have the following playback amplifier outputs:

3.6.3.2.1 Power amplifier output.- The output of the power amplifier shall be one watt across a 4 ohm resistive load when tested in accordance with 4.3.6.2. The output of this amplifier shall drive the internal loudspeaker and shall be available at the output jack.

3.6.3.2.1.1 Output jack.- An output jack, type JJ-034 in accordance with Specification MIL-J-641, shall be mounted on the front panel of the equipment. The amplifier output at the output jack shall be capable of driving a 600 ohm headset. When the headset plug is inserted in the jack, the internal loudspeaker shall be disconnected and the output signal shall be fed to the output jack.

3.6.3.2.2 Line amplifier output.- The output from each individual playback line amplifier shall be zero dbm when tested in accordance with 4.3.6.1. Internal level controls may be used for adjustment of the line output level. There shall be one line output available on the equipment terminal board for each of 4 channels.

3.6.3.3 Equalization.- Equalization shall be provided for each channel. The equalization shall be to NAB (National Association of Broadcasters) standard equalization curve for 7-1/2 inches per second tape speed.

3.6.3.4 Test record level.- A test record level shall be established by the contractor. This test record level shall be defined as the recording head current at 1000 cycles at which the recorder will meet the other requirements of this specification.

3.6.3.5 Distortion.- The total harmonic distortion shall not exceed 5 percent root mean square (r.m.s.) when tested as specified in 4.3.6.

3.6.3.6 Level control linearity.- For equal angles of rotation, the attenuation provided by all level controls shall be essentially equal in db throughout at least 80 percent of the control rotation.

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3.6.3.7 Amplifier construction.- Modular construction shall be used where possible. Amplifier and oscillator components shall be mounted on circuit boards using type C lead terminations in accordance with Standard MIL-STD-275. Each circuit board shall be easily removable to facilitate servicing. Guide rails shall be used to position the circuit boards in position, and each board shall be positively retained in its position.

3.6.4 Overall frequency response.- The overall frequency response shall be between two limits, one 8 db below the other. At a tape speed of 7-1/2 inches per second, the upper and lower limits shall be uniform from 100 to 7500 c.p.s.

3.6.5 Crosstalk.- When recording at test record level, the maximum crosstalk between any two channels shall be 35 db.

3.6.6 Signal-to-noise ratio.- The signal to overall noise ratio shall be at least 40 db when the signal level is reproduced at not greater than 5 percent harmonic distortion and measured against the noise level in accordance with 4.3.10.

3.6.7 Mode selector switches.- Each pushbutton or the rotary switch assembly (see 3.4.6) shall be capable of withstanding 100,000 actuations without degradation in performance when tested in accordance with 4.3.12.2.

3.6.8 Loudspeaker.- A permanent magnet, circular type loudspeaker having a nominal diameter of 2 inches shall be provided as an integral part of each recorder-reproducer. The output level of the loudspeaker when measured 2 feet on axis shall be at least 88 db (ref. 0.0002 dynes/cm^2) at a frequency of 400 c.p.s. with an input of 1 watt to the speaker. The loudspeaker shall have a frequency response of plus or minus 7 db ref. 1000 cycle signal between the limits of 500 to 5000 cycles when the output level is measured as specified in 4.3.12.1.

3.6.9 Supply line voltage and frequency variation.- The requirements of this specification with regard to tape motion and speed need not be met during voltage and frequency variations, but the machine shall continue to operate and shall not be damaged by these extremes.

3.6.10 Radio interference.- This equipment shall conform to the requirements of portable electric equipment in accordance with Specification MIL-I-17623.

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3.7 Physical durability. -

3.7.1 Exposure to extreme temperatures. - The equipment shall not suffer any permanent degradation in performance as the result of exposure to temperatures of from minus 20° to plus 131°F.

3.7.2 Resistance to temperature and humidity. - Temperature and humidity cycling, as performed in 4.3.13 shall neither render the equipment inoperative, cause a permanent degradation in performance, nor cause corrosion which would lead to degradation in performance, or cause other damage.

3.7.3 Life. - The recorder-reproducer shall be capable of 500 hours continuous operation when tested as specified in 4.3.14.

3.7.4 Rough handling. - All parts of the equipment shall be designed to have no critical adjustments such that normal rough handling and shipment will necessitate readjustment.

3.8 Painting. - The painting requirements of Specification MIL-I-983 shall apply only to exposed metallic surfaces.

3.9 Designation and marking. - In addition to the designation and marking requirements of Specification MIL-I-983, all controls and jacks shall be permanently marked to indicate function.

3.10 Technical manuals. - Technical manuals shall be furnished in accordance with Specifications MIL-I-983 and MIL-M-15071, except that the distribution requirement of Specification MIL-M-15071 is not applicable.

3.11 Accessories. - The following accessories shall be supplied:

(a) Low impedance dynamic microphone
(b) One standard commercial alignment tape used in final check out
(c) One 7 inch reel (empty)
(d) Test circuit boards or test cables

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4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General procedures.- The sampling procedures for the various tests to be performed on the equipment shall conform to 4.2.3. (The attention of the manufacturer is invited to the applicable paragraphs of Specification MIL-I-983. for information concerning these tests such as where conducted, failures and test conditions.)

4.2.1 Preproduction inspection.- Preproduction inspection shall be conducted at a laboratory satisfactory to the bureau or agency concerned (see 6.1). One sample of the magnetic tape recorder-reproducer covered in the specification shall be subjected to the following examination and tests which shall be performed in general in the order listed:

Test	Requirement paragraph	Test paragraph
General examination	3.3	MIL-I-983
Operating	----------	4.3.1
Supply line voltage and frequency variation	3.3 and 3.6.9	MIL-I-983
Shielding and radio frequency noise	3.6.10	MIL-I-17623
Airborne audible noise	3.3	MIL-I-983
Structureborne noise (vibration) for submarine equipment	3.3	MIL-I-983
Temperature	3.3 and 3.7.1	MIL-I-983
Temperature and humidity	3.7.2	4.3.13
Life	3.7.3	4.3.14

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Test	Requirement paragraph	Test paragraph
Vibration	3.5.3.1	MIL-I-983
Shock	3.5.3.1	MIL-I-983
Inclination	3.5.3.1	MIL-I-983
Rough handling	3.7.4	4.3.15
Loudspeaker	3.6.8	4.3.12
Enclosure	3.3	MIL-I-983
Tape speeds	3.4.3, 3.4.6.5, and 3.4.6.6	4.3.2
Mechanical requirements	3.5	4.3.4
Input impedance	3.6.3.1.2	4.3.5
Level control linearity	3.6.3.6	4.3.11
Flutter and wow	3.4.8	4.3.3
Output power and distortion	3.6.3.2.1, 3.6.3.2.2 and 3.6.3.5	4.3.6
Frequency response	3.6.4	4.3.7
Equalization	3.6.3.3	4.3.8
Crosstalk	3.6.5	4.3.9
Signal-to-noise ratio	3.6.6	4.3.10

4.2.1.1 Acceptance - preproduction inspection. - The results of each test shall be compared with the specification requirements. In the event of failure to conform to this specification, the contractor shall correct the cause of failure in the preproduction samples and on future production units.

4.2.2 Acceptance inspection. - The equipment selected in accordance with 4.2.3 shall be subjected to the following examination and tests in the order listed:

Test	Requirement paragraph	Test paragraph
Group A		
General examination	3.3	MIL-I-983
Operating	—	4.3.1

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Test	Requirement paragraph	Test paragraph
Group B		
Flutter and wow	3.4.8	4.3.3
Output power and distortion	3.6.3.2.1, 3.6.3.2.2 and 3.6.3.5	4.3.6
Frequency response	3.6.4	4.3.7
Equalization	3.6.3.3	4.3.8
Crosstalk	3.6.5	4.3.9
Signal-to-noise ratio	3.6.6	4.3.10
Group C		
Supply line voltage and frequency variation	3.3 and 3.6.9	MIL-I-983
Airborne audible noise	3.3	MIL-I-983
Structureborne noise (vibration) for submarine equipment	3.3	MIL-I-983
Shielding and radio frequency noise	3.6.10	MIL-I-17623
Tape speeds	3.4.3, 3.4.6.5, and 3.4.6.6	4.3.2
Mechanical requirements	3.5	4.3.4
Input impedance	3.6.3.1.2	4.3.5
Level control linearity	3.6.3.6	4.3.11
Loudspeaker	3.6.8	4.3.12
Enclosure	3.3	MIL-I-983
Temperature	3.3 and 3.7.1	MIL-I-983
Temperature and humidity	3.7.2	4.3.13
Vibration	3.5.3.1	MIL-I-983
Shock	3.5.3.1	MIL-I-983
Inclination	3.5.3.1	MIL-I-983
Life	3.7.3	4.3.14
Rough handling	3.7.4	4.3.15

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4.2.3 Sampling for acceptance inspection. -

4.2.3.1 Inspection lot. - All equipment presented for acceptance inspection at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.2.3.2 Sampling for group A inspection. - All equipment covered on the contract or order shall be subjected to the group A inspection listed in 4.2.2 with acceptance based on 4.2.3.5.

4.2.3.3 Sampling for group B tests. - A sample of equipments shall be selected from each inspection lot in accordance with table I and shall be subjected to each of the group B tests listed in 4.2.2 with lot acceptance based on sampling requirements of Standard MIL-STD-105.

Table I - Sampling for group B tests.

Number of equipments in inspection lot	Number of equipments in sample	(Number of equipments nonconforming on any group B test)	
		Acceptance number	Rejection number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.2.3.4 Sampling for group C tests. - Group C tests will be required when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be subjected to the group C tests listed in 4.2.2 with acceptance based on 4.2.3.7.

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4.2.3.5 Acceptance, group A inspection. - The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group A test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.2.3.6 Acceptance - group B tests. - The results of each test shall be compared with specification requirements. Failure to conform to the specification requirements for any test shall be counted as a defect and the equipment shall be rejected. If the number of such nonconforming equipments in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

4.2.3.7 Acceptance - group C tests. - The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.3 Test procedures. -

4.3.1 General inspection and operating tests. - The general inspection shall be conducted in accordance with Specification MIL-I-983. The operating tests shall consist of the following:

4.3.1.1 Code signals shall be recorded through the line input to channel 1 with the recorder-reproducer operating at 7-1/2 i.p.s. The signals shall then be replayed at a slower speed through the internal speaker, and the output jack. This test shall be made with the code reproduction capability.

4.3.1.2 Voice shall be recorded through the supplied microphone into the front panel microphone jack. The signals shall then be replayed through the internal speaker.

4.3.1.2.1 Signals shall be recorded on the remaining 2 channels through the line input. The signals shall then be replayed through the internal speaker.

4.3.1.3 All electrical and mechanical controls shall be actuated during this test to determine functional conformance with the requirements of this specification.

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4.3.1.4 The obtainable speeds shall be observed to determine compliance with the requirements of this specification.

4.3.2 Tape speeds.- All the tape speeds shall be measured to determine conformance with 3.4.3, 3.4.6.5 and 3.4.6.6.

4.3.3 Flutter and wow.- The equipment shall be tested for conformance with 3.4.8 by using a Furst meter or any other method satisfactory to the bureau or agency concerned. This test shall be run at 3-3/4 i.p.s.

4.3.4 Mechanical requirements.- The mechanical parameters shall be measured to determine conformance with 3.5.

4.3.5 Input impedance.- The line input impedance for each channel shall be measured to determine conformance with 3.6.3.2.2.

4.3.6 Output power and distortion.- A signal generator having an output distortion content of less than 0.3 percent r.m.s. shall be used as an input signal source. Input signals of 0.25 volt and 15 volts shall be recorded with the record level control adjusted to give a record head current at 1000 c.p.s. 6 db below test record level on all four channels.

4.3.6.1 Playback line amplifiers.- A 1000 c.p.s. signal shall be recorded and reproduced at 7-1/2 i.p.s. through each playback line amplifier with the amplifier output terminated into a 600 ohm resistive load. The output power of each amplifier shall be measured to determine conformance with 3.6.3.2.2.

4.3.6.2 Power amplifier.- A 1000 c.p.s. signal shall be recorded and reproduced at 3-3/4 i.p.s. and the output level control shall be adjusted to give one watt output across a 4 ohm resistive load. Frequencies of 100, 1000, 3000 and 5000 c.p.s. shall be recorded and reproduced retaining the output level setting. Distortion on all four channels shall be measured to determine conformance with 3.6.3.5. This procedure shall be repeated at 7-1/2 i.p.s. with an additional frequency of 7500 c.p.s.

4.3.7 Frequency response.- Input signals of 100, 1000, 3000, 5000 and 7500 c.p.s. at an input voltage of 0.25 volts shall be recorded at 7-1/2 i.p.s. and that record level specified in 3.4.6. These frequencies shall be reproduced with the output level control adjusted to give 1 watt output across a 4 ohm resistive load. The frequency response for each channel shall be measured across the 4 ohm resistive load to determine conformance with 3.6.4.

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4.3.8 Equalization.- A tape recorded on this equipment made up of frequencies of 100, 1000, 3000 and 5000 c.p.s. shall be played back on an equipment with known equalization characteristics. The measured results shall be corrected for the known characteristics and the corrected results be used to determine conformance with 3.6.3.3.

4.3.9 Crosstalk.- Frequencies of 100 and 5000 c.p.s. shall be recorded on channel 2 line input at test record level. The recording shall be reproduced with the output level control adjusted to give 1 watt output across a 4 ohm resistive load. The signal output of channels 1, 3 and 4 shall be measured across the 4 ohm resistive load to determine conformance with 3.6.5. This procedure shall be repeated, recording on channel 3, to determine channels 1, 2 and 4 conformance with 3.6.5.

4.3.10 Signal-to-noise ratio.- A 1000 cycle signal shall be recorded at test record level on all four channels. The recording shall be reproduced with the output level control adjusted to give 1 watt output across a 4 ohm load and the signal output level of each channel shall be measured across the 4 ohm resistive load. The 1000 c.p.s. recording shall be rerecorded with no signal input and the test procedure repeated to measure the noise level. This level shall be related to the signal level to determine conformance with 3.6.6.

4.3.11 Level control linearity.- The linearity of each level control shall be measured for conformance with 3.6.3.6.

4.3.12 Loudspeaker.- The loudspeaker shall be removed from the recorder and set up under free-field conditions. It shall be tested as follows:

4.3.12.1 Output level.- A 400 cycle signal of 1 watt shall be applied to the voice coil of the loudspeaker. The output level shall be measured 2 feet on axis, when the speaker is mounted in a simulated infinite baffle to determine conformance with 3.6.8.

4.3.12.2 Frequency response.- Signals between 500 and 5000 cycles shall be applied to the voice coil of the loudspeaker with an input of 1 watt while the loudspeaker is mounted in a simulated infinite baffle. Measurements shall be taken 2 feet on axis to determine conformance with 3.6.8.

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4.3.13 Temperature and humidity cycling.- This test shall consist of recording and reproducing radio program or other suitable electrical signals while the equipment is subjected at least four times to the following temperature and humidity cycle. At least 6 hours of this test shall consist of recording. The equipment shall be subjected to a temperature of plus 90°F. with a relative humidity of 90 percent for a period of 18 hours. The equipment shall then be immediately subjected to a temperature of plus 40°F. with a relative humidity of 90 percent for a period of 6 hours. The hot and cold cycle shall immediately begin again for a total of 4 cycles. Immediately after the fourth cycle, the equipment shall be given an operational test to determine that it is functioning properly and it shall be examined for corrosion and other damage. A recheck of the operational test, output distortion, frequency response, signal-to-noise ratio and flutter and wow tests shall be made 24 hours after the fourth cycle to determine conformance with 3.7.2.

4.3.14 Life test.-

4.3.14.1 Recorder-reproducer.- The recorder-reproducer shall be operated continuously for 500 hours in the following manner:

(a) Record code signals or single frequency at 7-1/2 i.p.s. for 50 hours.
(b) Reproduce the recorded code or single frequency at 3-3/4 i.p.s. for 200 hours using the code reproduction capability.
(c) Record voice or program information at 7-1/2 i.p.s. for 125 hours.
(d) Reproduce the recorded voice or program at 7-1/2 i.p.s. for 125 hours.

Upon completion of this test, the equipment shall be retested again for conformance with the operational test and flutter and wow test.

4.3.14.2 Mode switching.- The mode selector switch or switches from the equipment or an identical switch assembly submitted by the contractor shall be tested while removed from the equipment. The rotary switch or each push-button (see 3.4.6) shall be actuated 100,000 times. The switch assembly shall then be installed in the recorder and the equipment shall be retested to determine conformance with 4.3.1.

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4.3.15 Rough handling.- The equipment packed for normal shipment shall be held 24 inches above a concrete floor and dropped as follows:

(a) Once on a bottom corner.
(b) Once on the opposite top corner.
(c) Once on the bottom face.

The equipment shall then be unpacked and the operating test specified in 4.3.1 performed to determine conformance with 3.7.4.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Basic equipment.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical and machanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.- The onboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C respectively in accordance with Specification MIL-E-17555.

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5.2 Domestic shipment and storage or overseas shipment (see 6.1).

(5.2.1 Preservation, packaging, packing and marking.- The equipment and accessories, repair parts and technical publications shall be preserved and packaged levels A or C; packed levels A or B as specified (see 6.1) and marked in accordance with Specification MIL-E-17555).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Quantity of items required.
(c) Quantity of technical manuals required (see Specification MIL-M-15071).
(d) Supply demand control point for stock repair parts (see Specification MIL-P-15137).
(e) Laboratory where preproduction inspection shall be conducted.
(f) Level of packaging, packing, and marking required (see 5.1 and 5.2).
(g) If rack mounting adapter is required (see 3.5.3).

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the Unites States Government thereby incurs no responsibility or any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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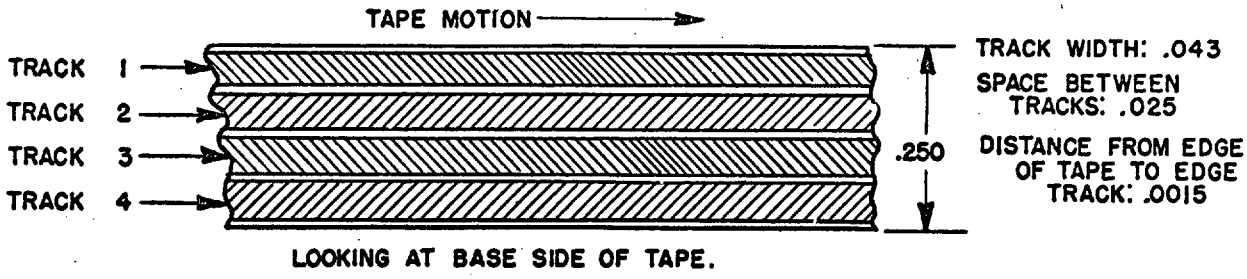


FIGURE 1--RECORDINGS DIMENSIONS

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SPECIFICATION ANALYSIS SHEET

Instructions

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D. C.

Specification

Organization	City	State

Contract No.

Quantity of Items Procured	Dollar Amount $

Material procured under a direct Government contract ☐ or a subcontract ☐

1. Has any part of the specification created problems or required interpretation in procurement?
   a. Give paragraph number and wording.

   b. Recommendations for correcting the deficiencies.

2. Comment on any specification requirement considered too rigid.

3. Is the specification restrictive? ☐ Yes ☐ No
   If the answer is "Yes", in what way?

4. Remarks (Attach any pertinent data which may be of use in improving this specification.)
Place this form and papers in an envelope and send to the Bureau.

Submitted by (Print name and activity)	Date

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